



COURSE PROJECT REPORT

Spring 2026

AI EXAM PROCTORING SYSTEM

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A project report submitted in partial fulfillment of the Requirements for the award of the degree of Bachelor of science in software engineering/Computer Science.

DEPARTMENT OF COMPUTER SCIENCE

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SALIM HABIB UNIVERSITY, KARACHI

DECLARATION

I/We hereby declare that this project report is based on my/our original work except for citations and quotations which have been duly acknowledged. I/We also declare that it has not been previously and concurrently submitted for any other degree or award at Salim Habib University or other institutions.

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APPROVAL FOR SUBMISSION

I/We certify that this project report entitled “AI Exam Proctoring System Using Java, JavaFX, and OpenCV” was prepared by (Amna Mustafa Alwani / Anosh Ashraf) has met the required standard for submission in partial fulfilment of the requirements for the award of Bachelor of Science at Salim Habib University.

Approved by,

Ms Sadia Karim

ACKNOWLEDGEMENTS

I/We would like to thank everyone who contributed to the successful completion of this project. Special thanks to our supervisor, Ms. Sadia Karim, for her guidance, support, and valuable suggestions throughout the development of this project.

We would also like to thank our parents, friends, and classmates for their encouragement and motivation during the project development process.

Yours Sincerely,

Student 1: ANOSH ASHRAF

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ABSTRACT

The rapid growth of online education has increased the need for secure and reliable examination systems. Traditional online examinations often face issues such as cheating, lack of supervision, and difficulty maintaining academic integrity. This project presents an AI Exam Proctoring System developed using Java, JavaFX, and OpenCV to monitor students during online examinations.

The system uses Artificial Intelligence and Computer Vision techniques to observe student activities in real time through webcam surveillance. It performs face detection, eye detection, and head movement tracking to identify suspicious behavior such as looking away from the screen, leaving the camera frame, or multiple face appearances.

Whenever suspicious activity is detected, the system automatically generates alerts, stores screenshots, records logs, and creates PDF reports for evidence and analysis. JavaFX provides an interactive graphical user interface while OpenCV handles image processing and detection.

The results demonstrate that the proposed system can effectively support secure online examinations, reduce cheating opportunities, and improve transparency in digital education environments.

APPENDICES

Folder Structure

ExamCheatingDetectionSystem/

|

|— src/

|

|— com.project.main/

|

|— com.project.report/

|

|— logs/

|

|— screenshots/

|

|— reports/

|

|— lib/

|

|— resources/

INTRODUCTION

Background

Online learning platforms and remote education systems have become increasingly popular worldwide. However, maintaining fairness and honesty during online examinations remains a major challenge. Unlike traditional classrooms, online examinations often lack direct supervision, creating opportunities for cheating and academic dishonesty.

To address this issue, the AI Exam Proctoring System was developed as an intelligent monitoring solution. The system uses webcam-based monitoring along with Artificial Intelligence and Computer Vision techniques to analyze student behavior during examinations.

Using OpenCV and Java technologies, the system can monitor facial visibility, eye movement, and head movement in real time. The system also captures screenshots and logs suspicious activities automatically.

Objectives

- Monitor students during online examinations using webcam surveillance
- Detect suspicious activities using Artificial Intelligence and Computer Vision
- Reduce cheating and improve academic integrity
- Generate automatic screenshots and logs as evidence
- Create PDF reports for examination review
- Support institutions in conducting secure online assessments

Problem Statement

During online examinations, institutions face several challenges:

- Students looking away from the screen
- Multiple people appearing in camera view
- Students leaving the examination screen
- Lack of evidence for suspicious behavior
- Difficulty maintaining fair examination environments

Therefore, an automated intelligent monitoring system is required to ensure transparency and fairness during online examinations.

Scope of Project

The project focuses on desktop-based real-time student monitoring using webcams. The system includes:

- Face detection
- Eye detection
- Head movement tracking
- Screenshot collection
- Logging system
- PDF report generation

The system is developed using Java and JavaFX and can be enhanced further with cloud integration and advanced AI technologies.

PROJECT METHODOLOGY

System Initialization

The system initializes JavaFX components and loads OpenCV libraries required for image processing.

Camera Activation

The webcam is activated using OpenCV VideoCapture to capture live video frames.

Frame Capture

The application continuously captures frames from the webcam during the examination.

Face Detection

Haar Cascade classifiers are used to detect faces from the webcam feed.

Features

- Detects student presence
- Detects face absence
- Tracks facial position
- Draws face boundary rectangle

Eye Detection

The system checks whether both eyes are visible.

Features

- Detects eye movement
- Monitors attentiveness
- Identifies hidden eyes

Head Movement Detection

The application monitors horizontal movement of the student's face.

Features

- Detects excessive movement
- Identifies suspicious looking-around behavior
- Maintains movement counter

Suspicious Activity Detection

The system generates alerts when suspicious behavior exceeds threshold limits.

Types of Alerts

- No face detected
- Eyes not visible
- Looking around
- Multiple face detection

Evidence Storage

Whenever suspicious behavior is detected:

- Screenshots are automatically stored
- Log records are generated
- Evidence is saved with timestamps

PDF Report Generation

The system generates detailed PDF reports containing:

- Violation summaries
- Activity logs
- Screenshot references
- Student monitoring details

TECHNOLOGIES USED

Technology	Purpose
Java	Main programming language
JavaFX	GUI and application interface
OpenCV	Real-time computer vision processing
Haar Cascade Classifiers	Face and eye detection
iText PDF	PDF report generation
IntelliJ IDEA	Development IDE
Webcam	Live video input

SYSTEM FEATURES

Real-Time Monitoring

The system continuously monitors students during examinations.

Features

- Live webcam streaming
- Continuous frame processing
- Real-time monitoring

Screenshot Evidence Collection

The application automatically stores screenshots whenever suspicious activity occurs.

Features

- Timestamp-based image saving
- Automatic evidence collection
- Organized screenshot storage

Logging System

All suspicious activities are stored in log files.

Features

- Event tracking
- Date and time recording
- Activity history maintenance

PDF Report Generation

Detailed examination reports are generated using the iText PDF library.

Features

- Violation summaries
- Student activity logs
- Screenshot references
- Examination monitoring reports

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Purpose of the System

The purpose of this system is to:

- Prevent cheating during online examinations
- Improve examination transparency
- Monitor students automatically
- Support secure online education systems

Functional Requirements

User Monitoring

The system shall:

- Monitor students using webcam surveillance
- Detect faces and eyes
- Track suspicious head movement

Alert Generation

The system shall:

- Generate alerts for suspicious activities
- Notify when face is not visible
- Detect multiple faces

Evidence Collection

The system shall:

- Save screenshots automatically
- Store activity logs
- Generate PDF reports

Non-Functional Requirements

Performance Requirements

- The system should process webcam frames efficiently
- Real-time detection should occur without major delays

Reliability Requirements

- Face detection should remain accurate during execution
- Logs and screenshots should be stored correctly

Usability Requirements

- The interface should be simple and user-friendly
- Alerts should be easy to understand

Maintainability Requirements

- Code should be modular and properly documented
- Future upgrades should be easy to implement

Hardware Requirements

- Webcam or Laptop Camera
- Minimum 4GB RAM
- Intel i3 Processor or higher
- 500MB Free Storage

Software Requirements

- Java JDK 17
- JavaFX SDK

- OpenCV Library
- IntelliJ IDEA
- Windows Operating System

SOFTWARE DESIGN SPECIFICATION (SDS)

User Interface Module

Handles the graphical interface and displays webcam monitoring.

Face Detection Module

Uses OpenCV Haar Cascade classifiers to detect student faces.

Eye Detection Module

Analyzes eye visibility and movement.

Movement Analysis Module

Tracks head movement and suspicious behavior.

Logging Module

Stores suspicious activity records and timestamps.

Output Module

Displays alerts and generates PDF reports.

System Workflow

Input → Webcam Activation → Face Detection → Eye Detection → Movement Analysis → Alert Generation → Screenshot Saving → PDF Report Generation

IMPLEMENTATION DETAILS

System Architecture

Input Layer

- Webcam video feed
- Live student activity

Processing Layer

- Face detection module
- Eye detection module
- Movement analysis module
- AI behavior analysis
- Logging system

Output Layer

- Status alerts
- Screenshots
- Logs
- PDF reports

Algorithms Used

Haar Cascade Classifier

The project uses Haar Cascade classifiers for detecting faces and eyes.

Advantages

- Fast detection
- Lightweight processing
- Real-time performance

Sample Suspicious Activities Detected

- No face detected
- Eyes not visible
- Looking around repeatedly
- Multiple faces detected
- Student leaving camera view

Testing

Feature	Status
Camera Detection	Passed
Face Detection	Passed
Eye Detection	Passed
Screenshot Saving	Passed
Logging System	Passed
PDF Report Generation Passed	

Results

The system successfully:

- Detected faces and eyes in real time
- Monitored suspicious movement
- Generated alerts for abnormal behavior
- Stored screenshots and logs
- Generated PDF reports successfully

CHALLENGES FACED

During the development process, several challenges were encountered:

- Maintaining detection accuracy under different lighting conditions
- Managing real-time webcam processing efficiently
- Handling multiple suspicious activity scenarios
- Synchronizing GUI updates with real-time monitoring
- Storing screenshots and logs without delays

FUTURE ENHANCEMENTS

Future versions of the system may include:

- AI emotion detection
- Voice activity monitoring
- Cloud database integration
- Student authentication system
- Mobile application support
- Online admin dashboard
- Machine learning-based behavior prediction
- Multi-camera monitoring system
- Integration with LMS platforms

SDG GOALS ALIGNMENT

SDG 4 – Quality Education

The system supports fair and transparent examinations, helping institutions improve educational quality.

SDG 9 – Industry, Innovation and Infrastructure

The project applies Artificial Intelligence and Computer Vision technologies to modernize online examination systems.

SDG 16 – Peace, Justice and Strong Institutions

The system promotes fairness, accountability, and trust in educational institutions.

CONCLUSION

In conclusion, the AI Exam Proctoring System successfully achieves its objective of monitoring students during online examinations using Artificial Intelligence and Computer Vision technologies.

The system provides:

- Real-time webcam monitoring
- Face and eye detection
- Suspicious activity identification
- Automatic screenshot and log generation
- PDF report creation

The project improves transparency, fairness, and trust in online education systems while reducing opportunities for cheating.

REFERENCES

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THANK YOU